

MIMIC NLP

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Course: AI in Healthcare



Introduction

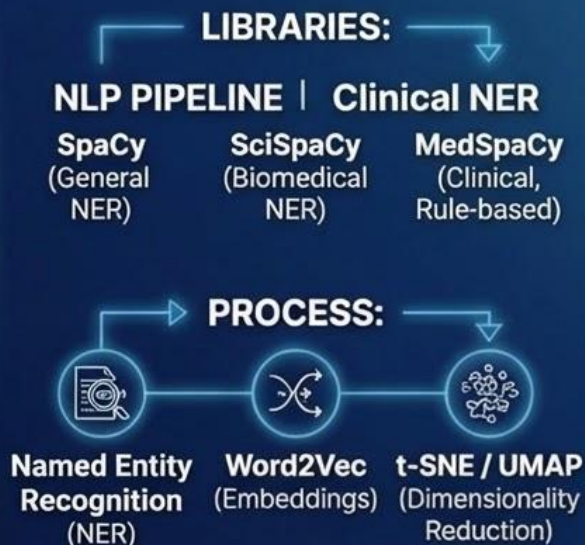


GOAL & TARGET

- Extract medical entities from MIMIC-III discharge notes and visualize relationships.
- **Target:** Acute Kidney Injury (AKI) — ICD-9 code 584.9.
- AKI is a sudden decline in renal function, one of the most common and costly in-hospital complications.



NLP PIPELINE LIBRARIES



ADVANCED & VISUALIZATION

- **BONUS:** ClinicalBERT (Contextual) vs. Word2Vec (Static).
- **DISPLACY:** Visualize inline entity annotations.
- Compare entity extraction, nearest neighbors, and best t-SNE across models.

DATA OVERVIEW: MIMIC-III NOTEEVENTS



SOURCE & SCALE

- Source: MIMIC-III **NOTEEVENTS** table (~2 million rows, >1 GB compressed).
- Data read in chunks of 50,000 rows for memory efficiency.



**2M+
ROWS**



FILTERING & SELECTION

- Filtered to Discharge Summary category.
- **AKI Keywords:** 'acute kidney injury', 'acute renal failure', 'acute tubular necrosis', 'atn', 'renal insufficiency', 'dialysis', 'crrt'.
- Selected 50 notes for analysis (manageable runtime).



**50
NOTES**



NOTE CHARACTERISTICS

- Average note **length:** ~10,000+ characters per note (long, detailed clinical documents).
- **Text preprocessing:** Stripped MIMIC de-identification brackets **[** **]** and collapsed excessive whitespace.



SpaCy: Named Entity Recognition



MODEL & STATS

- **Model:** en_core_web_sm (trained on general English web text, OntoNotes corpus).
- Extracts entities across 18 label types.
- DisplaCy visualization confirms broad, non-clinical labeling scheme.



TOP ENTITY TYPES

- 5 **CARDINAL:** Lab values, vitals, dosages.
-  **ORG:** Often misclassified medical terms.
-  **PERSON:** Physician/patient names.
-  **DATE:** Admission/discharge, procedures.
-  **GPE:** Place names in patient histories.



INTERPRETATION

- General-purpose: Identifies entities well but not tuned for medical vocabulary.
- Misclassified clinical terms (e.g., drug names tagged as ORG).
- Serves as a baseline for comparison.

SpaCy: Word2Vec, t-SNE & UMAP



WORD2VEC EMBEDDING



- Trained on 50 tokenized notes.
- Embedding dimension: 100.
- Obtained embeddings for entity representations using averaged word vectors.



VISUALIZATION (t-SNE & UMAP)

t-SNE (TUNED PERPLEXITIES)

- **Perplexity 5:** Very tight, small local clusters (spotting near-duplicates).
- **Perplexity 15:** Local structure begins to emerge.
- **Perplexity 30:** Balanced view (global structure visible, local preserved).
- **Perplexity 50:** Broader groupings, fine-grained local detail lost.

UMAP (n_neighbors: 5, 15, 30)

- Preserves both local and global structure better than t-SNE.
- **At n_neighbors=15:** Clear separation between entity types (DATE, CARDINAL, ORG clusters).



INTERPRETATION

- Embeddings show numbers/lab values cluster together; date-related terms form their own group.
- General entities (ORG, PERSON) are more scattered as the model was not trained on clinical text.

SciSpaCy: Named Entity Recognition



MODEL & STATS



- Model: en_core_sci_sm (trained on biomedical literature, PubMed abstracts).
- Finds significantly more entities than standard SpaCy due to biomedical training.



KEY OBSERVATIONS



- **Medical Terms:** Identifies drugs ('metoprolol', 'furosemide'), diagnoses ('acute kidney injury'), and anatomy ('renal cortex').
- **Limitation:** A single 'ENTITY' label prevents direct distinction between types (drugs vs. diseases vs. anatomy) without extra processing.
- **Vocabulary:** Biomedical vocabulary is inherent in the model from scientific text training.



INTERPRETATION



- SciSpaCy is more aggressive at entity detection in clinical text.
- The single entity label necessitates downstream categorization (e.g., UMLS linking) for fine-grained analysis.

SciSpaCy: Word2Vec, t-SNE & UMAP



WORD2VEC: BIOMEDICAL EMBEDDINGS



- Higher coverage than SpaCy due to biomedical tokenization, capturing more clinical terms.
- Biomedical tokenizer preserves terms like 'creatinine' and 'hemodialysis' as single tokens.



VISUALIZATION (t-SNE & UMAP)



t-SNE (PERPLEXITIES: 5, 30, 50)

- **Perplexity 5:** Very granular clusters (individual drug families, specific anatomy regions).
- **Perplexity 30:** Natural groupings emerge (medications, anatomy, lab values form dense regions).
- **Perplexity 50:** Broad category separation visible across the projection.

UMAP (BETTER STRUCTURE)

- Shows tighter, more interpretable clusters than t-SNE at equivalent settings.
- Drug names, lab tests, and anatomical terms form distinct islands.



INTERPRETATION & IMPACT



- Because SciSpaCy extracts genuine biomedical entities, Word2Vec embeddings capture meaningful clinical relationships.
- Similar drugs appear near each other, and related conditions cluster together.
- Ideal for feature engineering in downstream ML tasks.



MedSpaCy: Named Entity Recognition



APPROACH & AKI RULES



- Method: Rule-based extraction (TargetMatcher, custom TargetRules).
- Categories: PROBLEM, MEDICATION, PROCEDURE, TEST.
- AKI Rules: Renal conditions, nephrotoxic drugs, procedures, lab tests.



ENTITY BREAKDOWN (EXAMPLES)



PROBLEM

'acute kidney injury'
'hyperkalemia'
'sepsis'
'hypertension'



MEDICATION

'furosemide'
'vancomycin'
'insulin'
'heparin'



PROCEDURE

'hemodialysis'
'CRRT'
'renal biopsy'
'intubation'



TEST

'creatinine'
'BUN'
'GFR'
'urinalysis'



INTERPRETATION: PRECISION vs. RECALL



- Extracts fewer entities than SciSpaCy (strict rule adherence).
- Highly precise, clinically meaningful, and correctly categorized.
- Demonstrates a valuable precision-over-recall trade-off for clinical NLP.

MedSpaCy: ConText Algorithm

MedSpaCy's unique feature: detecting clinical context modifiers around entities.



CONTEXT DETECTION RESULTS



-  **NEGATED:** e.g., 'no fever', 'denies chest pain' — entities flagged as not present.
-  **UNCERTAIN:** e.g., 'possible pneumonia' — entities flagged as uncertain.
-  **HISTORICAL:** e.g., 'history of hypertension' — entities flagged as past conditions.
-  **FAMILY:** e.g., 'father had diabetes' — entities attributed to family members.
-  **HYPOTHETICAL:** e.g., 'if seizures recur' — entities flagged as conditional.
-  **AFFIRMED (present & active):** Remaining entities confirmed as current and present.



WHY THIS MATTERS: CLINICAL IMPACT



- In clinical NLP, simply detecting a mention of a condition like 'renal failure' is insufficient.
- ConText specifies whether the patient currently **HAS** the condition, **DOESN'T** have it, or had it in the **PAST**.
- This is critical for building accurate clinical features and downstream models.
- No other tested library provides this level of context-awareness out of the box.

MedSpaCy: Word2Vec, t-SNE & UMAP



EMBEDDINGS & MODEL



- MedSpaCy reuses the SpaCy-based **Word2Vec** model (no native tokenizer).
- Reducing coverage: Many multi-word entities absent from Word2Vec vocabulary.



VISUALIZATION RESULTS (t-SNE & UMAP)

- **t-SNE** (Perplexities 5, 15, 30, 50):
 - Fewer data points, but clear categorical separation.
 - Perplexity 15-30 is optimal for smaller dataset.
- **UMAP**:
 - Cleanest separation of three libraries.
 - Four clinical categories form visually distinct clusters.



INTERPRETATION: INTERPRETABILITY



- Despite fewer entities, visualizations are highly interpretable.
- Every entity is clinically meaningful.
- Labeled categories provide natural color-coding, revealing embedding space organization of clinical concepts.

Side-by-Side Comparison



KEY INSIGHT: Each library serves a different purpose: SpaCy provides breadth, SciSpaCy provides biomedical coverage, and MedSpaCy provides clinical precision with context awareness.



SpaCy



Approach:
Statistical (ML)



Training Data:
Web text



Entity Types:
18 labels



Context Detection:
No



Clinical Specificity:
Low



SciSpaCy



Approach:
Statistical (ML)



Training Data:
Biomedical lit.



Entity Types:
1 label (ENTITY)



Context Detection:
No



Clinical Specificity:
Medium



MedSpaCy



Approach:
Rule-based



Training Data:
N/A (rules)



Entity Types:
4 clinical labels



Context Detection:
Yes (ConText)



Clinical Specificity:
High

Entity Overlap Analysis



OVERLAP STATISTICS



- **SpaCy & SciSpaCy:** Significant overlap (both ML-based), but SciSpaCy detects significantly more biomedical terms.
- **SpaCy & MedSpaCy:** Low overlap, confirming they target distinct entity types (general vs. clinical).
- **SciSpaCy & MedSpaCy:** Moderate overlap where the biomedical model identifies rule-defined clinical terms.
- **All Three:** A small core set of universally recognized medical concepts.



WHAT THIS TELLS US

- **Distinct Spaces:** Low overlap between SpaCy and MedSpaCy confirms separate target entities (general vs. clinical).
- **SciSpaCy as a Bridge:** SciSpaCy shares entities with both, linking general and specific clinical terminology.



PRACTICAL RECOMMENDATION



- **Optimal Clinical NLP Pipeline:** Combine SciSpaCy (for broad biomedical entity detection) and MedSpaCy (for context-aware clinical extraction).
- **Best Coverage & Accuracy:** This combination ensures both breadth (SciSpaCy) and clinical precision (MedSpaCy).
- **For AKI specifically:** SciSpaCy broadly catches renal terminology, while MedSpaCy precisely identifies and contextualizes conditions, medications, and procedures related to Acute Kidney Injury.

t-SNE Tuning: Effect of Perplexity

Tested perplexity values: 5, 15, 30, 50

LOW PERPLEXITY (5)



- Focuses on very local neighborhoods (5 nearest neighbors).
- Produces many small, tight clusters — good for finding near-duplicate or highly similar entities.
- Can fragment natural groups into many small pieces.

MEDIUM PERPLEXITY (15-30)



- Best balance for most datasets — shows both local structure and emerging global patterns.
- At perplexity 30, entity categories start to separate visibly.
- This is the recommended default range for exploratory analysis.

HIGH PERPLEXITY (50)



- Considers more of the global structure.
- Large, broad groups visible, but fine-grained local detail is smoothed out.
- Useful for seeing the "big picture" of how entity categories relate overall.

KEY TAKEAWAY

- There is no single best perplexity — tuning it reveals different aspects of the data.
- Presenting multiple perplexities side-by-side gives the most complete view.

UMAP: Complementary to t-SNE



UMAP ANALYSIS

- Run alongside **t-SNE** for each library.
- Key parameter: **n_neighbors** (tested at 5, 15, 30), analogous to perplexity.



ADVANTAGES OVER t-SNE

- Better preserves **global structure** (clusters far apart remain far apart).
- **Significantly faster** computation, especially on larger datasets.
- **More deterministic** results across runs (less sensitivity to initialization).



OBSERVATIONS FROM OUR RESULTS

- UMAP produced **cleaner, more separated clusters** for all libraries.
- Particularly effective for **MedSpaCy's 4 clinical categories** (PROBLEM, MEDICATION, PROCEDURE, TEST).
- For **SciSpaCy**, revealed **subclusters** within the broad ENTITY label.



KEY TAKEAWAY

- UMAP is a **strong complement** to t-SNE. While t-SNE excels at revealing local structure, UMAP provides a **more faithful representation of the overall topology** of the embedding space.

Entity Extraction Comparison Across Models

Comparing how each model handles specific AKI-relevant clinical terms



TERMS COMPARED (12 AKI TERMS)

- hypertension
- diabetes
- creatinine
- dialysis
- metoprolol
- furosemide
- sepsis
- pneumonia
- acute kidney injury
- renal failure
- insulin
- anemia



SciSpaCy

- Excels at **broad biomedical vocabulary** (drugs, diseases, lab values).
- Highest **recall** for clinical terms due to biomedical training.



MedSpaCy

- Precise for **specific, rule-defined AKI terms**.
- Adds critical context: **negation, uncertainty**.
- Best **precision** and context for defined terms.



SpaCy

- Misses most clinical terms (trained on general web text).
- Rarely recognizes terms like 'creatinine' or 'furosemide'.
- Not suitable as a **standalone clinical NLP tool**.



KEY TAKEAWAY

- For AKI entity extraction, **SciSpaCy** provides the best **coverage**, while **MedSpaCy** provides the best **precision + context**.

Word2Vec Nearest Neighbors Comparison

Comparing semantic neighborhoods across SpaCy vs SciSpaCy Word2Vec models



PROBE TERMS & METHODOLOGY

5 key AKI entities:
creatinine, dialysis,
hypertension, sepsis, insulin

- Retrieved top-10 nearest neighbors for each term from both models.
- Compared neighbor lists and measured overlap.



KEY FINDINGS

- SciSpaCy tokenization produces **more clinically meaningful** neighbors (e.g., drugs, lab values) due to biomedical term preservation.
- SpaCy tokenization often reflects **morphological similarity** (subwords) rather than semantic relatedness.
- Low overlap indicates fundamentally **different representations** of clinical text between models.



EXAMPLES & ANALYSIS

- **'creatinine'**: SciSpaCy neighbors include **related lab values (BUN, GFR)**; SpaCy may include unrelated tokens.
- **'dialysis'**: SciSpaCy captures **related procedures (hemodialysis, CRRT)** and conditions.
- **'insulin'**: SciSpaCy neighbors are **diabetes-related**; SpaCy may return **dosage-related** terms.

Best t-SNE Visualization: Top AKI Entities

SciSpaCy t-SNE at perplexity=30, filtered to top 50 most frequent entities



WHY THIS IS THE BEST t-SNE



SciSpaCy model: Captures the most **clinically relevant** entities, providing the richest set of biomedical terms.



Perplexity=30: The standard sweet spot that balances **local cluster tightness** with **global separation**.



Filtered to top 50 entities: Removes **noise** from rare entities, making patterns clearly visible.



Every point is **labeled** with its entity text for easy reading.



WHAT THE VISUALIZATION SHOWS



Meaningful clusters where related medical concepts group together (e.g., renal terms, medications, lab values).



Confirms that Word2Vec trained on clinical text learns useful **semantic relationships**.



Examples: '**creatinine**', '**BUN**', and '**potassium**' cluster near each other, as do 'furosemide', 'heparin', and 'insulin'.



Clear separation between entity types validates the embedding quality for downstream tasks.

[BONUS] ClinicalBERT: Setup & Results

Leveraging contextual embeddings from MIMIC-III notes for enhanced clinical NLP



MODEL & CONFIGURATION

- **Model:** emilyalsentzer/Bio_ClinicalBERT (Pre-trained on MIMIC-III notes).
- Produces **contextual embeddings** (vs. static Word2Vec), meaning the same word gets different vectors based on context.
- Embedding dimension: **768** (vs. 100 for Word2Vec).
- Generated embeddings for entities using the **[CLS] token**.



t-SNE & UMAP RESULTS

- Clusters are **tighter and more coherent** than Word2Vec-based projections.
- Related medical concepts previously separated are now **grouped together**.
- The **768-dimensional space** captures richer semantic relationships.



CLINICALBERT vs. WORD2VEC

- ClinicalBERT shows **more meaningful clustering** of clinical entities.
- **Context-dependency** handles ambiguity (e.g., 'discharge summary' vs. 'nasal discharge').
- **Trade-off:** ClinicalBERT is **much slower** and requires more compute than Word2Vec.

Key Takeaways



INTEGRATED LIBRARIES & CONTEXT



No Single Library Does It All: Combining **SpaCy** (breadth), **SciSpaCy** (biomedical depth), and **MedSpaCy** (clinical precision) yields the best results.



Context Matters in Clinical NLP: MedSpaCy's **ConText algorithm** is essential; a significant percentage of entities were **negated**, highlighting the risk of incorrect clinical conclusions without context.



VISUALIZATION TECHNIQUES (t-SNE & UMAP)



Tuning t-SNE is Critical: Presenting **multiple perplexity values** is necessary to show the full structure of the embedding space. **Perplexity=30** is the best single choice for AKI entity analysis.



UMAP Complements t-SNE: UMAP better preserves **global structure** and is faster. Using **both** provides the most complete understanding of entity relationships.



EMBEDDINGS & DOWNSTREAM ML



Contextual > Static Embeddings: **ClinicalBERT** is the state-of-the-art for clinical NLP embeddings because it captures **context**, resulting in **tighter, more meaningful clusters**.



Clinical NLP Enables Better ML: Extracted entities and their embeddings can serve as **features for downstream ML models** (e.g., predicting **patient outcomes, readmission risk, AKI severity staging**).

Conclusion



FOCUS & DATA PIPELINE

- **PROJECT FOCUS:** Full clinical NLP pipeline on **MIMIC-III** discharge notes for **Acute Kidney Injury (AKI)** – ICD-9 code 584.9.
- **Data & Preprocessing:** Extracted discharge summaries from NOTEVENTS, filtered for AKI keywords, cleaned text.
- **Entity Extraction:** **SpaCy** (intention community (general)), **SciSpaCy** (biomedical), **MedSpaCy** (clinical, rule-based).



EMBEDDINGS & ANALYSIS

- **Word2Vec Embedding:** Trained on tokenized notes, generated entity embeddings.
- **Dimensionality Reduction:** Tuned **t-SNE** (4 perplexity values) & **UMAP** (3 n_neighbors values).
- **Context-Aware Analysis:** **MedSpaCy's ConText** algorithm for negation, uncertainty, historical, family, hypothetical.
- **ClinicalBERT Contextual Embeddings** (Bonus): Comparison to Word2Vec.



KEY RESULTS

- Each library has distinct strengths: SpaCy gives a broad overview, **SciSpaCy** excels at biomedical term detection, and **MedSpaCy** provides clinically precise, context-aware extraction.
- Entity extraction comparison showed **SciSpaCy** as the **best for AKI term coverage**; **MedSpaCy** for **precision + context**.
- Word2Vec neighbors confirmed **SciSpaCy** learns more **clinically meaningful** semantic relationships.
- **Best t-SNE:** **SciSpaCy** at perplexity=30, filtered to top 50 entities - reveals **clear clinical clustering**.